Using Prime Alpha to Separate Skill from Luck

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The active management industry is facing an existential crisis. Subpar returns in active equity strategies over the last decade have led to massive outflows from active strategies into passive strategies. Management fees have declined precipitously as lower-cost smart beta strategies encroach on the domain long held by active managers. Investors are questioning the value-add of active management as well as their own abilities to select the best performing strategies.

In one sense, the inability of active managers to outperform the market indexes shouldn't be surprising. After all, numerous studies have suggested that the average manager performs in line with the market before fees and as a result, underperforms the market after fees. This result is consistent with the notion that in totality, all the assets managed by the asset managers in the industry add up to be the market and therefore cannot outperform or underperform itself. Two studies, Malkiel [1995] and Fama and French [2010], suggest active managers in aggregate do not outperform their benchmarks after fees.

The past decade has been particularly difficult for active managers. During the Global Financial Crisis, investors were disappointed by the absolute and relative drawdowns they experienced in their investment portfolios. To make matters worse, since the Global Financial Crisis, the percentage of managers outperforming their benchmarks has plummeted (see Exhibit 1). Indeed, the median active fund has underperformed its benchmark after fees in each of the last seven years. These results are troubling for an industry that charges fairly high fees for its active management services.

With this as the backdrop, asset owners, portfolio managers, and risk managers are looking for ways to improve their odds of success. This quest has become increasingly important as future asset returns are expected to be meager given low interest rates and elevated equity prices. Asset owners and investors want better techniques to identify skill in active managers. Portfolio managers on their end are looking for ways to demonstrate their value-add and to differentiate their services. Finally, risk managers aspire for tools that monitor investment strategies to ensure they are performing as expected and that allow them to identify deviations from processes.

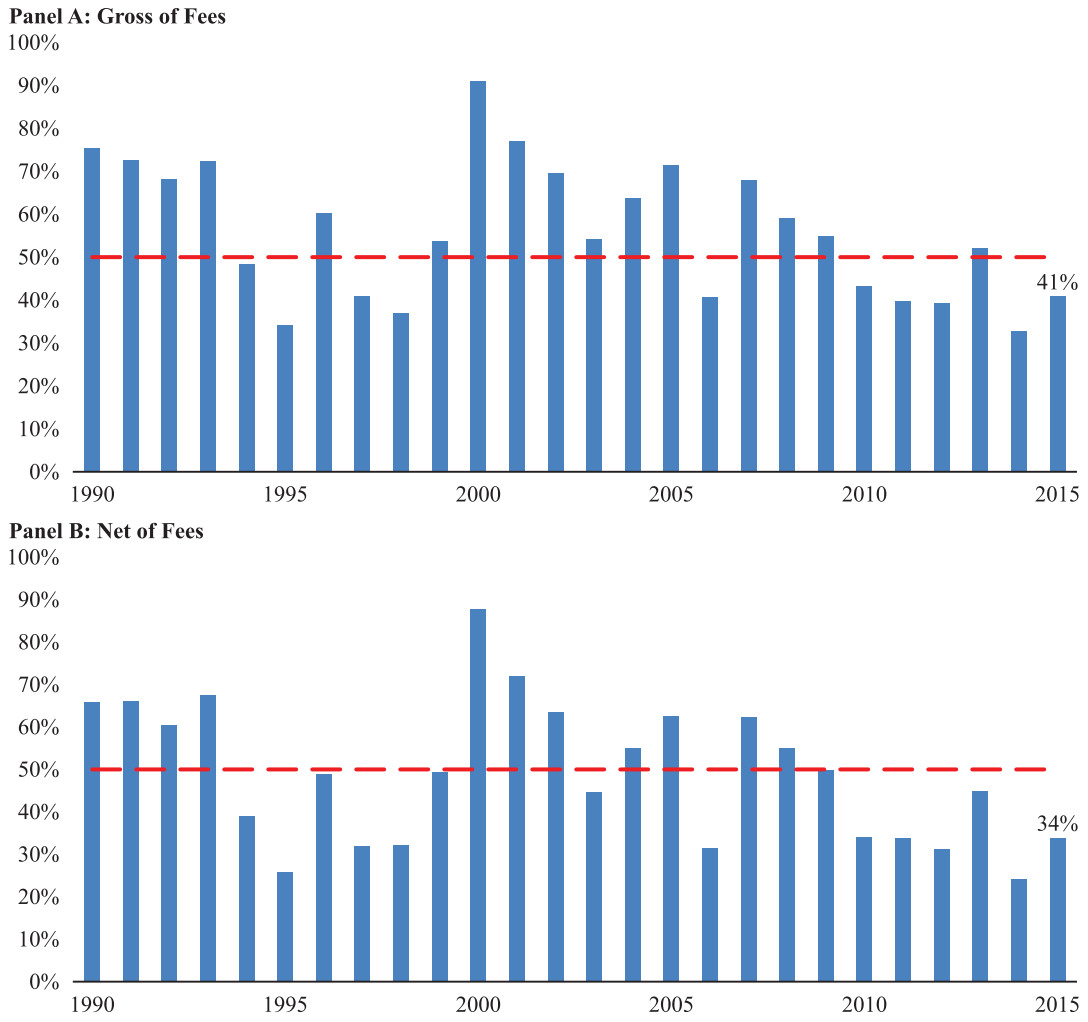
In Chin and Gupta [2017], we introduced *prime alpha* as a metric that separates skill from luck in fixed-income strategies. In this article, we extend those findings to a broad range of equity strategies and show that prime alpha is also an important indicator of skill in these universes.

Many prior studies on active management skill have used the Morningstar or CRSP (Center for Research in Security Prices) databases of mutual fund returns to

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EXHIBIT 1

Percentage of U.S. Large-Cap Strategies Outperforming Their Respective Benchmarks



Source: eVestment.

make conclusions about the largely retail funds of the asset management industry. In addition, other studies have made use of Informa Investment Solutions' PSN Investment Manager Database to assess the returns of separately managed accounts. In theory, it would be ideal to collect the investment results of all the market participants, but to date, no such database exists.

By focusing on mutual funds, the databases commonly used in literature omit a significant segment of the market—namely, the large institutional investors who use standard and customized strategies directly with asset managers. These institutional investors comprise more than 40% of the global assets under management in 2012 according to a study by PriceWaterhouseCoopers [2014] and are almost as large as the retail investors.

To study the strategies used by institutional asset owners, we use the data from eVestment, which is an independent company that maintains the premier database for institutional asset managers. We find that the eVestment database covers more assets and strategies than other sources and, as a result, enhances the representativeness of the market participants in our study. In addition, a vast majority of asset owners and consultants around the world use and vet this database to select managers to oversee client assets. Thus, even though asset managers self-report their returns to eVestment,

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they have an incentive to submit their information accurately so that they can be included in potential searches for new mandates. Other researchers have also scrutinized the eVestment database. For example, Jenkinson, Jones, and Martinez [2016] stated that "constant scrutiny from clients using this data guarantees a high degree of accuracy." Finally, we compared our net-of-fees outperformance rates to S&P Global's SPIVA scorecards and find that they are similar, especially after 2000.[1] These quality checks ensure that the eVestment dataset is robust and representative even though the returns are self-reported.

We make four contributions to the existing literature. First, we analyze the persistence of manager returns using a unique dataset that tracks the returns of the largest asset owners in the world. Many studies have used the returns of U.S. mutual funds to assess manager skill, but we use a database that contains the returns to strategies that are commonly used by pension funds, sovereign wealth funds, endowments, and other institutional clients. Second, we extend the analysis period to the end of 2015, thus capturing the Global Financial Crisis and its recovery. Including this period is important because active managers were tested during the downturn as well as during the recovery. Third, unlike prior studies, we include the two new factors—robust minus weak and conservative minus aggressive—from Fama and French [2014] in the regressions to incorporate the latest research on the drivers of stock returns. Finally, most of the previous studies have focused on active managers within the U.S. large-cap equity universe. In our study, we extend the analysis to include funds invested in U.S. small-cap, global, and international equities by creating universe-specific factors for each of the categories. These extensions and enhancements are rare in literature because the datasets are not readily available.

DATA

eVestment's global manager database contains about 10,000 firms and more than 41,000 long-only vehicles as of 2015. We apply various filters and adjustments to this universe to create our research database.

First, we focus on equity funds classified as "active" by eVestment because we are interested in studying the skill of active managers. Within each of the universes we are studying (U.S. large-cap equities, U.S. small-cap equities, global equities, and international equities), we select those funds with benchmarks that are commonly used in those categories. Then, even though the dataset contains monthly returns from the 1960s, we focus on the period from 1990 to 2015 to ensure there are sufficient observations within each of the different manager universes.

We also collapse strategies with more than one vehicle into one single strategy by taking the average of the returns across all the vehicles. Firms may distribute their strategies through different vehicles and share classes depending on the needs of their clients. Our adjustment mitigates potential biases in overweighting certain strategies in the dataset.

After applying these filters and adjustments, we are left with approximately 6,200 strategies in our dataset. Exhibit 2 contains the annual strategy counts as well as the number of strategies entering and leaving the dataset. Interestingly, strategies did not leave the dataset until the early 2000s, suggesting there may be survivorship bias prior to 2000 because we would expect some funds to cease operations during the 1990s. Although Jenkinson, Jones, and Martinez [2016] did not find any survivorship bias in the eVestment dataset, we will assess the potential impact of survivorship bias in our dataset by looking at different subperiods to determine if our results differ.

In this article, we use gross-of-fee returns because our focus is on the skill exhibited by active managers, regardless of the fee they charged for this skill. In our view, manager skill should capture the outcomes of the investing decisions made by the portfolio manager, not the marketing and distribution decisions made by the firm. Gross-of-fee returns capture our intentions nicely. It is also important to note that in practice, investors can use our framework to identify those managers with skill as well as the magnitude of their value-add. They can then assess whether that skill justifies the fees paid to the manager. We believe this transparency provides the investors with more knowledge when engaging with portfolio managers on the level of fees they are willing to pay.

Some funds in the eVestment database do not have gross-of-fee returns even though they show net-of-fee returns. In these instances, we estimate the gross-of-fee return by adding the net-of-fee return to the median fee from the funds with both sets of fees within a manager universe.

EXHIBIT 2
eVestment Database

	Fund Counts											
	U.S. Large Cap			U.S. Small Cap			Global			International		
Year	Start	Left	New	Start	Left	New	Start	Left	New	Start	Left	New
1990	248	0	70	88	0	24	26	0	8	46	0	14
1991	318	0	61	112	0	28	34	0	14	60	0	21
1992	379	0	83	140	0	36	48	0	18	81	0	19
1993	462	0	102	176	0	47	66	0	12	100	0	16
1994	564	0	86	223	0	46	78	0	20	116	0	41
1995	650	0	110	269	0	40	98	0	22	157	0	27
1996	760	0	114	309	0	66	120	0	14	184	0	39
1997	874	0	118	375	0	69	134	0	17	223	0	32
1998	992	0	127	444	0	93	151	0	17	255	0	47
1999	1119	0	165	537	0	72	168	0	26	302	0	35
2000	1284	0	196	609	0	63	194	0	27	337	0	46
2001	1480	11	236	672	1	80	221	0	37	383	1	57
2002	1705	25	178	751	10	77	258	4	25	439	11	32
2003	1858	39	218	818	22	54	279	10	40	460	18	40
2004	2037	68	166	850	28	81	309	2	44	482	15	36
2005	2135	65	161	903	28	66	351	5	64	503	12	58
2006	2231	100	166	941	34	51	410	9	58	549	13	47
2007	2297	101	159	958	40	50	459	6	64	583	17	58
2008	2355	155	99	968	72	56	517	20	69	624	31	48
2009	2299	148	69	952	75	42	566	14	50	641	29	28
2010	2220	164	70	919	73	30	602	29	51	640	32	27
2011	2126	173	59	876	53	21	624	37	47	634	44	18
2012	2012	163	41	844	54	17	634	35	40	609	30	19
2013	1890	185	38	807	46	22	639	53	30	598	46	10
2014	1743	120	17	783	49	10	616	42	9	562	34	19
2015	1640	125	–	744	41	1	583	33	–	547	33	1

Source: eVestment.

The eVestment database provides the assets under management (AUM) covered by the strategies. These figures are self-reported by the asset managers, and we found that the quality of this data is much better in the recent period. At the end of 2015, the AUM covered by the strategies in our study sample were approximately $4.5 trillion, $0.7 trillion, $1.6 trillion, and $1.8 trillion within the U.S. large-cap, U.S. small-cap, global, and international universes, respectively. We believe this asset coverage is sufficiently large across the universes and, therefore, our dataset is representative of the strategies used in the industry.

To decompose the manager returns, we use a set of factors that are not yet widely used in literature, although we expect that to change. Most studies have used Fama–French's [1993] three factors and Carhart's [1997] momentum factor to decompose returns. Following Fama–French's 2014 article, which added two new factors to their model, we use market beta, size, value, robust minus weak, conservative minus aggressive, and price momentum as the factors in our analysis. We think these six factors will be commonly used in asset pricing models going forward. For the U.S. large-cap study, we obtain the monthly factor returns from the Fama–French (beta, small minus big, value minus growth, robust minus weak, conservative minus aggressive) and Carhart (momentum) websites.

For the studies on U.S. small-cap, global, and international strategies, we calculate similar variations of the same factors in the respective universes to make

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EXHIBIT 3
Factor Construction

Panel A: Factor Definitions

Factor	Definition	Rationale
Beta	Underlying benchmark	Market exposure
Size	Small minus big	Exposure to small-cap stocks
Value	Value minus growth	Exposure to cheap stocks
Robust minus Weak	Top minus bottom return on equity	Exposure to high profitability stocks
Conservative minus Aggressive	Bottom minus top asset growth	Exposure to low asset growth stocks
Momentum	Top minus bottom trailing 12-month momentum	Exposure to high price momentum stocks
U.S. Dollar Movements*	DXY	Exposure to currency movements against the USD
Emerging Markets*	MSCI Emerging Markets Index	Exposure to global emerging markets

Panel B: Universes and Methods Used to Construct the Factor Returns

Universe	Stock Universe	Factor Construction
U.S. Large Cap	Largest U.S. stocks	Fama–French/Carhart factors
U.S. Small Cap	Russell 2000 stocks	Top–bottom quintiles
Global	MSCI Global Developed Market stocks	Top–bottom quintiles
International	MSCI EAFE & Canada stocks	Top–bottom quintiles

**Used in the regressions on the global and international market manager universes.*

the analyses more relevant and comparable. Exhibit 3 provides an overview of our data sources and methods. Because many of the factors outside the U.S. large-cap equity universe are not commonly used in literature, we calculate our own monthly factor returns to ensure there is consistency in their construction. For each of the universes, we perform the following steps:

1. Create a dataset without survivorship bias containing companies from the constituents of the underlying universe.
2. Every month, form quintiles for a factor by first sorting by that factor and then putting the same number of names within each quintile.
3. Calculate quintile returns by equally weighting the returns of the stocks in each quintile.
4. Define the factor return as the return of the top quintile minus the return of the bottom quintile.
5. Repeat steps 2 to 4 for all the factors.

We use quintiles in our factor construction rather than follow the exact formulations used by Fama and French because quintile studies are very common in the industry and are easily replicable. In addition, robustness checks indicate that the results do not change using the slightly different construction method.

RETURN DECOMPOSITION

We use a standard returns-based technique to analyze fund returns. We first define active return as the difference between the gross return of the portfolio and that of its benchmark. We make this calculation using monthly data.

$$AR_{i,t} = R_{i,t} - B_{i,t}, \tag{1}$$

where $AR_{i,t}$ is the relative return of strategy i in month t and is calculated as the difference between $R_{i,t}$, the absolute return of strategy i in month t, and $B_{i,t}$, the monthly return of strategy i's benchmark during month t.

We then decompose the active return $AR_{i,t}$ by running a regression using the Fama–French [2014] factors and the Carhart [1997] momentum factor.

$$AR_{i,t} = \alpha_i + \beta_i(MARKET_t - RF_t) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + m_i MOM_t + \varepsilon_i, \tag{2}$$

where $MARKET_t$ is the return of the relevant benchmark during month t; RF_t is the U.S. three-month T-bill in month t; SMB_t, HML_t, RMW_t, and CMA_t are the small minus big, value minus growth, robust

minus weak, and conservative minus aggressive factors, respectively, in month t; MOM_t is the momentum factor in month t; and ε_i is the residual from the regression.

In evaluating multi-country strategies, we run additional analyses to determine the impact of cross-country factors on portfolio returns. In addition to running Regression 2, we run another regression with two additional factors:

$$AR_{i,t} = \alpha_i + \beta_i(MARKET_t - RF_t) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + m_i MOM_t + f_i FX_t + w_i(EM_t - MARKET_t) + \varepsilon_i, \quad (3)$$

where the new variables FX_t represents the return of a currency basket during month t and EM_t represents the return of MSCI Emerging Markets (in USD) during month t. The other variables are defined as before.

Global and international strategies may manage currency exposures actively (currency positions are set deliberately with a view) or passively (currency positions are determined solely by the underlying investments), but the return impact will be captured in portfolio returns. For example, a USD-based strategy that does not hedge its non-USD exposures will experience a return pattern that is tied explicitly to the performance of the U.S. dollar. To account for those instances where currency is not part of the investment decision, we use the DXY as a proxy for currency movements. The DXY measures the value of the U.S. dollar relative to its most important trading partners. We think DXY is a good proxy to capture the impact of currency decisions in investment portfolios because the U.S. dollar is widely traded and investors use it as a cross currency when adjusting for exposures to non-USD exposures.

Managers of multi-country strategies may have structural overweights or underweights to the emerging market countries depending on their mandate. Companies in emerging market countries may offer higher returns over time, but these higher returns typically come with more risk. As a result, we introduce another factor to capture the strategic impact of these decisions. Note that the emerging markets factor used in the regression is the incremental return above and beyond the market factor—this is to capture the specific return from investing in emerging markets. This factor captures the strategic and persistent allocation to emerging markets, not the tactical shifts into those areas.

The returns of the factors used in our analyses exhibit cyclicality—while there may be positive means in some of the return patterns, the factors ebb and flow depending on the prevailing market conditions. We will show that this cyclicality plays an important role in our persistence studies later.

PRIME ALPHA

We define *prime alpha* as the difference between the active return and the factor returns:

$$PA_{i,t} = AR_{i,t} - (\beta_i(MARKET_t - RF_t) + s_i SMB_t + h_i HML_t + r_i RMW_t + c_i CMA_t + m_i MOM_t). \quad (4)$$

An alternative interpretation of prime alpha is that it is the sum of the intercept and the residual terms from Regression 2. The residual is included because it captures the unexplained impacts of the portfolio manager's decisions, and this idiosyncratic component may be a key part of the manager's added value. In addition, there is an economic interpretation to the residual—it captures the portion of the returns not attributable to common factors and the intercept, but yet, it impacts the actual returns of the investors. Therefore, including the residual allows us to account for all the components of fund returns.

Our formulation of prime alpha is consistent with the industry's definition of manager skill. Since strategic and persistent exposures to factors are normally not attributable to skill, we exclude their impact in our calculation of prime alpha by including these exposures in the factor structure in Regression 2. In addition, since the returns from tactical factor bets as well as stock selection are considered a part of manager skill, we use prime alpha to capture these components by defining it to be the net impact after removing the returns from a series of common factors. As a result, prime alpha measures manager skill nicely.

We will now show prime alpha is persistent across time and can be used as an indicator of skill (or lack thereof).

PERSISTENCE OF PRIME ALPHA

We test return persistence by running regressions of future prime alphas against trailing prime alphas over different periods. Specifically, we look over trailing

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three- and five-year periods to predict the next three and five years, respectively, because these are common periods used by the industry to assess manager performance and manager skill.

For the three-year analysis, at time t,

$$PA_{i,t,t+35} = \alpha_i + \beta_i PA_{i,t-36,t-1} + \varepsilon_i, \quad (5)$$

where $PA_{i,t,t+35}$ is the prime alpha of strategy i calculated over the period t to $t + 35$ (i.e., next three years) and $PA_{i,t-36,t-1}$ is the prime alpha of strategy i calculated over the period $t - 36$ to $t - 1$ (i.e., last three years).

For comparison, we also show the results of the regressions using trailing active returns.

$$AR_{i,t,t+35} = \alpha_i + \beta_i AR_{i,t-36,t-1} + \varepsilon_i, \quad (6)$$

where $AR_{i,t,t+35}$ is the active return of strategy i calculated over the period t to $t + 35$ (i.e., next three years) and $AR_{i,t-36,t-1}$ is the active return of strategy i calculated over the period $t - 36$ to $t - 1$ (i.e., last three years).

Because we run these regressions every month, we have overlapping regressions. To correct for the biases in significance thresholds, we adjust the t-statistics to account for the fact that we have fewer observations than our pooled regressions suggest.

Analogous formulations were used for the five-year analysis.

RESULTS FOR U.S. LARGE-CAP STRATEGIES

Exhibit 4 shows the results of running Regressions 5 and 6. From Panel A, we see that trailing active returns are not persistent. These active returns do not control for exposures to common factors and therefore are subjected to the movements of these factors over time. Indeed, directionally, the coefficient of –0.05 suggests active returns are mean reverting. That is, managers with high trailing active returns tend to have low forward active returns. Contrary to industry practice, investors shouldn't choose funds through a screening process that uses active returns—that process would yield the opposite of the intended outcomes. This mean-reverting behavior is significant at the 1% level.

One simple enhancement is to focus on beta-adjusted returns by removing the impact of the market factor in active returns. Panel B of Exhibit 4 shows that the results are directionally better when the market factor is included in the regression. Indeed, there is some persistence in these market-adjusted active returns.

Panel C of Exhibit 4 shows that prime alpha is even more effective because it accounts for more factors that impact returns. Prime alpha is persistent with a coefficient of 0.12 and is statistically significant at the 1% level. These results on U.S. large-cap strategies are consistent with the findings in Iachini, Lam, and Peterson [2011], even though we use a broader set of factors.

EXHIBIT 4
U.S. Large-Cap Strategies, 1990–2015

Panel A: Forecast Forward Active Returns Using Trailing Active Returns

	3-Year Analysis	5-Year Analysis
Coefficient	–0.05	–0.09
t-Statistic of Coefficient	–4.90	–5.61

Panel B: Forecast Forward Active Returns (net of market factor) Using Trailing Active Returns (net of market factor)

	3-Year Analysis	5-Year Analysis
Coefficient	0.04	0.01
t-Statistic of Coefficient	3.84	0.82

Panel C: Forecast Forward Prime Alphas Using Trailing Prime Alphas

	3-Year Analysis	5-Year Analysis
Coefficient	0.12	0.13
t-Statistic of Coefficient	12.05	8.19

Panel D: Forecast Forward Active Returns Using Trailing Prime Alphas

	3-Year Analysis
Coefficient	0.03
t-Statistic of Coefficient	2.19

Panel E: Forecast Forward Prime Alphas Using Trailing Prime Alphas over Different Periods; 3-Year Analysis

	1990–2002	2003–2015
Coefficient	0.12	0.13
t-Statistic of Coefficient	6.59	10.24

Panel F: Prime Alpha Regressions Using Different Subsets of Data; 3-Year Analysis

	S&P 500	Russell 1000 Growth	Russell 1000 Value
Coefficient	0.12	0.09	0.03
t-Statistic of Coefficient	8.15	4.36	1.38
Number of Strategies	1613	662	630

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Our results are different from those found in Busse, Goyal, and Wahal [2010] (BGW). That study found little evidence of manager skill and alpha persistence in U.S. large-cap, mid-cap, and small-cap strategies. However, there are some significant differences between our study and that of BGW. First, the eVestment dataset contains a much broader set of strategies. For example, our dataset contains over 2,000 U.S. large-cap equity strategies at the end of 2008 (the last year of the BGW study), while BGW contained a little over 1,000 strategies in this cohort. Second, we use monthly returns instead of quarterly returns, so we are able to uncover deeper insights from the more granular data. Finally, and probably most importantly, our reading of BGW suggests the researchers assessed all the strategies by using large-cap factors as the independent variables. We improve on that method by applying factors that are calculated over the universe of stocks relevant to the specific strategy (i.e., the value factor used to assess skill in a small-cap strategy is derived from the stocks in the small-cap universe). As explained earlier, this makes the factors more comparable to the respective strategies we are studying. Taken together, our enhancements to the data and methodology account for our stronger findings on manager skill.

Barras, Scailliet, and Wermers [2010] (BSW) suggested that it is difficult to find managers with true skill because the statistical methods we use have margins of error. BSW also found that by using a higher threshold to determine statistical significance, it is possible to have more confidence in the abilities of those managers who are beyond that threshold. Our findings on the persistence of prime alpha are across all managers on average but consistent with BSW, we also find that the persistence results are stronger for those managers with the highest prime alpha. That is, the coefficient of the persistence parameter increases from 0.12 to 0.19 for the subset of managers in the top quartile ranked by prime alpha. Taken in totality, these studies suggest investors should focus on those strategies with the highest prime alpha to increase their odds of picking skilled managers.

A natural question that arises is why active returns are mean reverting. We believe one significant reason is that factor returns have cycles and exhibit strong patterns of mean reversion over rolling three- and five-year horizons. Because active returns include the impact from factors, they will also exhibit mean reversion. Many portfolios are exposed to factor returns, either strategically (per investment style or discipline) or tactically (through portfolio manager views on opportunities), and the subsequent returns from these exposures can overwhelm the returns from the skill portion of active portfolios. For example, a value investment strategy exposes an investor to the returns of the value factor. These returns can be substantial: We observed long stretches where the value factor performed extremely well and other periods where the factor struggled mightily. This means that over horizons where factor returns can dominate (say, over three to five years), fund returns will be heavily influenced by the returns of factors regardless of other decisions made by the portfolio managers. Because these factor returns are highly unpredictable, portfolio returns that are heavily influenced by factor returns may be a result of luck, while portfolio returns that are heavily influenced by idiosyncratic elements may suggest skill.

In practice, portfolio managers may not explicitly set the factor exposures of their strategies "top down." Nevertheless, these portfolios may still be exposed to factors as a result of decisions made at the stock level, whether intentional or not. In our experience, we find that "bottom-up" managers tend to believe the risks associated with their stock decisions are purely idiosyncratic but ex-post performance analyses would typically unearth the importance of common factors.

The persistence of prime alpha is compelling, but many investors do not experience pure prime alpha returns in their actual returns unless they hedge out all the factor exposures of their managers. As a result, we also study the efficacy of using prime alpha to predict forward active returns in Panel D of Exhibit 4. Intuitively, we expect the results of this regression to be weaker than those in Panel C because forward active returns will be influenced by the returns of factors as well as the returns from prime alpha, so we should see less persistence. Nevertheless, the results suggest that prime alpha predicts forward active returns, thus giving us confidence in a metric that can guide manager assessment and manager selection.

These findings have important implications for investors who wish to assess manager skill and the potential to add value going forward. Selecting managers by past raw returns will tend to produce subpar results, while ranking managers by prime alpha tends to

yield the intended outcomes. Our research shows that those managers with the highest prime alpha will tend to deliver more value-add in the future.

ROBUSTNESS CHECKS FOR U.S. LARGE-CAP STRATEGIES

To test robustness, we run the same analysis over different time periods and using different subsets of data. First, we compare the studies using five-year look-back and look-ahead periods in Exhibit 4 (Panels A, B, and C) to the studies using three-year windows. Generally, the results are very similar, but the statistical significance declines in the longer-horizon studies partly because of the smaller number of observations. It is also possible that over longer horizons, portfolio managers and investment processes may change and therefore it is more difficult to see persistence.

In addition, as noted earlier, we believe there may be survivorship bias in the dataset prior to the early 2000s. We therefore run the same prime alpha regressions for two subperiods (1990–2002 and 2003–2015) to determine whether our results were unduly influenced by the earlier period. Our findings, shown in Panel E of Exhibit 4, suggest that the conclusions are very similar—prime alpha is persistent in both halves of the data with similar coefficients.

Finally, we run the same analysis for different subsets of the dataset. In the results presented so far, we had pooled together all the U.S. large-cap managers into the same regression. We now analyze the results for the subsets of managers with similar investment philosophies: core (S&P 500 benchmark), growth (Russell 1000 Growth benchmark), and value (Russell 1000 Value benchmark). Our conclusions remain the same after studying those results in Panel F of Exhibit 4.

ANALYSIS AND RESULTS FOR U.S. SMALL-CAP, GLOBAL, AND INTERNATIONAL STRATEGIES

As shown in Exhibit 5, Panel A, the regression results for U.S. small-cap strategies suggest trailing active returns are not persistent. The result using rolling three-year periods is not statistically significant, but the rolling five-year result is significant—the negative coefficients on trailing active returns suggest that trailing active returns for small-cap funds are mean reverting, just like their large-cap counterparts.

EXHIBIT 5
U.S. Small-Cap Strategies, 1990–2015

Panel A: Forecast Forward Active Returns Using Trailing Active Returns

	3-Year Analysis	5-Year Analysis
Coefficient	–0.01	–0.06
t-Statistic of Coefficient	–0.80	–2.61

Panel B: Forecast Forward Prime Alphas Using Trailing Prime Alphas

	3-Year Analysis	5-Year Analysis
Coefficient	0.08	0.08
t-Statistic of Coefficient	5.59	3.65

The results in Panel B of Exhibit 5 using prime alpha are consistent with the U.S. large-cap studies. The coefficients are positive and statistically significant, implying that prime alpha is persistent among small-cap funds.

Unlike the vast literature we found on U.S. managers, there are only a few studies on the returns of non-U.S. strategies and the skill of those managers. This is probably due to a couple of factors—the data on non-U.S. funds is harder to obtain and there are difficulties in dealing with other return factors like currency movements. Tonks [2005] found persistence in fund alphas for pension fund returns in the United Kingdom over the period 1983–1997.

Exhibit 6 contains the regression results for return persistence on global funds. The results on active returns suggest that there is no persistence in this metric over three-year horizons, but it is not necessarily mean reverting like the U.S. evidence. Indeed, the low *t*-stat suggests that the coefficient is likely 0, implying that trailing results have no correlation to future returns. Consistent with the U.S. evidence, however, the results in Panel B still suggest there is persistence in prime alpha, thus demonstrating that factors play a prominent role in global portfolio returns and that factor cycles exist beyond the U.S. markets.

In the rolling five-year regression results, both active returns and prime alpha are persistent. The coefficient in the prime alpha regression is higher, suggesting that it is more economically significant.

Comparing the results of the regressions with and without the two additional factors in Panel C of Exhibit 6, we find there is little difference between the

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EXHIBIT 6
Global Strategies, 1990–2015

Panel A: Forecast Forward Active Returns Using Trailing Active Returns

	3-Year Analysis	5-Year Analysis
Coefficient	0.01	0.07
t-Statistic of Coefficient	0.31	2.87

Panel B: Forecast Forward Prime Alphas Using Trailing Prime Alphas

	3-Year Analysis	5-Year Analysis
Coefficient	0.06	0.11
t-Statistic of Coefficient	2.44	2.74

Panel C: Forecast Forward Prime Alphas Using Trailing Prime Alphas With Different Factors; 3-Year Analysis

	Six Factors	Eight Factors
Coefficient	0.05	0.06
t-Statistic of Coefficient	2.05	2.44

Notes: Eight-factor model includes market, size, value, profitability, asset growth, momentum, currency (DXY), and emerging markets (MSCI EM). The six-factor model excludes the last two factors.

EXHIBIT 7
International (global ex-U.S.) Strategies, 1990–2015

Panel A: Forecast Forward Active Returns Using Trailing Active Returns

	3-Year Analysis	5-Year Analysis
Coefficient	0.05	0.02
t-Statistic of Coefficient	2.38	0.56

Panel B: Forecast Forward Prime Alphas Using Trailing Prime Alphas

	3-Year Analysis	5-Year Analysis
Coefficient	0.10	0.11
t-Statistic of Coefficient	4.76	3.72

Panel C: Forecast Forward Prime Alphas Using Trailing Prime Alphas With Different Factors; 3-Year Analysis

	Six Factors	Eight Factors
Coefficient	0.05	0.10
t-Statistic of Coefficient	2.78	4.76

Notes: Eight-factor model includes market, size, value, profitability, asset growth, momentum, currency (DXY), and emerging markets (MSCI EM). The six-factor model excludes the last two factors.

six- and eight-factor models. Both models have statistically significant coefficients. Although our prior had been that the currency and emerging market factors should have significant impacts on manager returns, the empirical evidence does not bear this out. Nevertheless, it is still our view that manager skill should be assessed using these two additional factors because they can overwhelm manager skill over short periods of time.

Exhibit 7 shows the regression results for international (global ex-U.S.) strategies. Note that active returns and prime alpha both seem to be persistent when using three-year windows, but only prime alpha is persistent in the five-year study. In addition, the coefficients of the prime alpha regressions are larger and more economically significant.

Finally, the model using the additional two factors seems superior to the base model, as shown in Panel C of Exhibit 7.

These findings confirm the results on U.S. large-cap strategies. We see robust evidence that prime alpha is persistent across different strategy universes in the United States and outside the United States. Therefore, we believe that manager skill should be assessed using prime alpha, not active return.

CONCLUSIONS

The active management industry has been in turmoil since the Global Financial Crisis. Disappointed by the poor returns of active managers since the Global Financial Crisis, asset owners are moving assets from active strategies to passive strategies. Yet, given the low expected returns going forward, asset owners are looking for ways to enhance their returns. The asset managers themselves are also facing significant headwinds: They are struggling to demonstrate their value-add in light of increasing evidence that skill is rare and the fact that fees can overwhelm the benefits of active management.

With these challenges as the backdrop, prime alpha is a timely tool that gives investors confidence in their ability to select managers. Managers with high prime alpha are consistently able to deliver high active returns above and beyond the returns of their benchmarks and a series of common factors. Conversely, managers with low prime alpha are more likely to be unskilled and may only outperform their benchmarks if they happen to be exposed to factors that have strong concurrent performance.

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This research has significant implications for investors because they should focus on prime alpha when selecting managers for investment. Although other studies suggest true manager skill is rare and even strong observed alphas may be a result of luck, we find there is performance persistence across the whole manager universe. That is, high (low) prime alpha is generally followed by high (low) prime alpha. We also find there is stronger persistence among those strategies with the highest prime alpha. Prime alpha thus helps asset owners find the managers with the highest probability of success going forward.

This research is also important for asset management firms in differentiating the investment teams and investment strategies they have in-house. By applying prime alpha to separate those teams with true skill and those who were lucky, organizations can focus compensation and marketing resources on those teams with the greatest potential to add value going forward. In addition, prime alpha gives the portfolio managers a way to demonstrate their skills above and beyond factors that are increasingly available as low-cost options.

Finally, risk managers can use prime alpha to assess the value-add of portfolio managers and their adherence to their stated objectives by monitoring factor exposures over time.

We are very encouraged by our robust results across different asset classes and different manager categories. Combined with our findings across fixed-income strategies in Chin and Gupta [2017], our results give us confidence that prime alpha is a holistic framework that asset owners, portfolio managers, and organizations can apply to differentiate skill from luck.

ENDNOTES

We thank an anonymous referee for providing valuable feedback.

[1]SPIVA U.S. Scorecard Year-End 2015, www.spglobal.com/our-insights/SPIVA-US-Scorecard.html.

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